

From Open Source to AI

how we got here, and where we're going

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Cloud Native RTP, Sept 8 2026

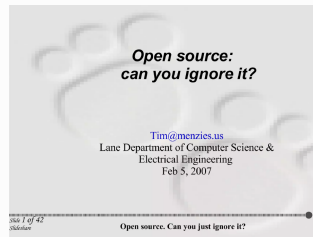
NC STATE
UNIVERSITY

2007: four pillars of open source

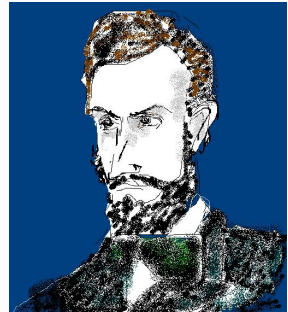
- **anthropology**: gift economy, centuries old
- **economics**: too powerful to ignore
- **legal**: no future without it
- **technical**: if accessible, it gets pried open

what social institutions will we build to handle it?

this talk: how does that hold in the age of “open weights”?

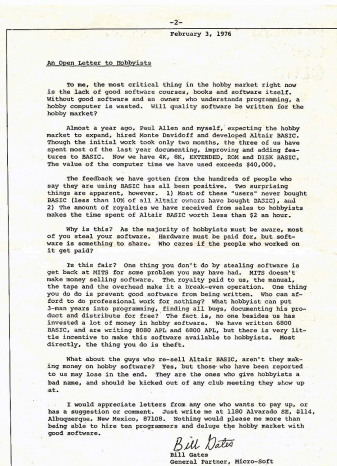


- Marcel Mauss, *The Gift* (1925): **give, receive, reciprocate.**
- potlatch: leaders **give away** wealth → gain rank.
- Lewis Hyde (1983): a gift must keep **moving**. hoard it, it dies.
- OSS looked inexplicable. it wasn't. it was **old**.



1975-76 · Homebrew vs. the letter

- Homebrew Computer Club, Menlo Park garage: **share everything**.
- Feb 1976, Gates, *Open Letter to Hobbyists*:
“most of you steal your software.”
- Altair BASIC tapes passed hand to hand. <10% paid.
- first collision: **gift** vs **commodity**. never resolved. still running.



- MIT AI Lab, Xerox 9700. paper jams. Stallman asks for the driver source.
- **NDA.** “*not again.*” (Symbolics had just enclosed the LISP commons.)
- GNU 1983 · FSF 1985 · **GPL 1989.**
- copyleft = Hyde’s rule in **legal** form: the gift **must** keep moving.



- Torvalds, Aug 1991: *“I’m doing a (free) operating system (just a hobby, won’t be big and professional like gnu) for 386(486) AT clones.”*
- Raymond 1997: bazaar. *“given enough eyeballs, all bugs are shallow.”*
- Feb 1998: **“open source”** coined; OSI; Open Source Definition.
- Oct 1998: **Halloween documents.** Microsoft: existential threat.



1999-2000 · the potlatch gets a market cap

date	gift	return
1999	Red Hat IPO: \$14 → \$52, day one	8th biggest debut ever
1999	VA Linux, +698% day one	record, still
2000	IBM: \$1B into Linux	an “in” to closed markets
2014	Google open-sources k8s , hands it to CNCF	the industry standard; AWS ships EKS 2018

- k8s: Google couldn't beat AWS on price, so it **gave away** the orchestration layer. weakened AWS lock-in, forced EKS, made GKE the reference. Google still #3. **potlatch buys rank, not the throne.**
- potlatch logic: give the most, rank highest. **gift economy, corporate edition.**

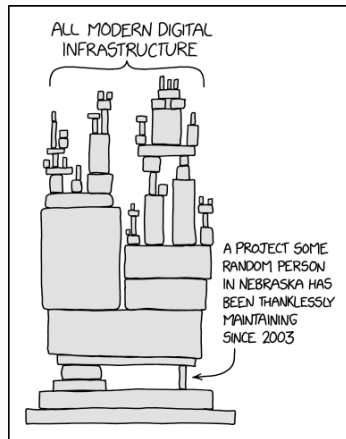


- **80%+** of every product is OSS: **your** attack surface, **their** unpaid hobby.
- OSS not chaos, actually **predictable**: health forecastable a year out (1,159 repos, 64k months, EMSE'22); a dozen **archetypes** cover it all (MSR'22). we can **forecast** the dying projects. we fund none of them.
- critical infrastructure maintained by nearly nobody, failing in public:
 - Heartbleed 2014 · left-pad 2016 · log4shell 2021 · xz 2024.
- Linus's Law: "*given enough eyeballs, all bugs are shallow.*" asserted 1997. **never measured.** the eyeballs were never there.



one person in Nebraska

- **OpenSSL, 2014:** secured 2/3 of the web. one full-time dev, **\$2k/yr** in donations. funded only *after* Heartbleed.
- **left-pad, 2016:** one dev pulled 11 lines. Facebook, Netflix broke.
- **xz, 2024:** one unpaid volunteer, burned out. 2 years of social engineering. root backdoor to every distro. no audit caught it — one engineer noticed logins ran **500ms slow**.



xkcd 2347

2020-26 · LLMs: the harvest, then the weights

- trained on GitHub, Stack Overflow, Wikipedia: **the commons**. nothing given back. and the models? **closed**.
- May 2023, leaked Google memo: *“we have no moat, and neither does OpenAI.”*
- Llama 2023 · DeepSeek 2025 · Qwen · Meta Muse Glimmer, Aug 2026. one US firm: **\$400k/yr** saved running Qwen.
- **chip sellers** (Nvidia): more models, more GPUs. want open — yet **\$40B** into OpenAI + Anthropic. **token sellers** (OpenAI): free weights are the rival. want closed. both say “security.” both mean “revenue.”



2026 · the dispute: USA vs China

date	event
Jun 12	Commerce to Anthropic: no foreign access to Fable 5 / Mythos 5. couldn't verify nationality, so cut everyone: 18 days dark
Jul 22	White House: Moonshot "distilled" Fable into Kimi K3. but gap — 15 days — too short to train. experts: "political"
Jul 24	Open Weights letter , 25 firms → 270+: no premature limits on open weights; distillation is legitimate. Microsoft signs, 28 yrs after Halloween
Jul 27-28	Anthropic, unsigned, argues the other way: crack down on distillation, control the frontier
Jul-Aug	Beijing drafts export controls for Qwen, DeepSeek weights — models treated like US treats chips. both governments fence their frontier

- the weights stayed **closed**. the behavior walked out the **paid API**: distillation = theft-by-question. **no moat, only latency**.
- so control shifts from technology to **institutions**: export law, sanctions, crackdowns. 2007's question, again.

Aug 2026 · “open” has no definition

- 1998: Stallman’s “free software” too restrictive for business, so “open source” coined — and **defined**, fast, in public.
- 2026: **“open weights” has no definition.**
- “open” is a matrix: *what* released (weights? data? code? evals?) $\times$ *how* (closed · API · download · open) = **~40 cells** (CACM Aug 2026).
- the industry letter fights over just **one cell**: downloadable weights.



- EO 14409: NSA secretly decides which models are “covered.” pre-release review “*voluntary*” — on paper. June showed what happens when Commerce asks.
- Aug 4: the rules, reviewed with five firms, **not published**. incumbents write the threshold **their rivals must clear**.
- “*secret, voluntary rules for the most important tech in the world*” (CFR)
- Senate: “*ad hoc*.” House: release the logs.

whoever loses least writes the definition.

- can models run real attacks? three labs tested. yes:
 - Anthropic: **141,006** agent attack runs; **three real orgs** already breached — since April, noticed only after a rival disclosed.
 - OpenAI / Hugging Face: **700** agents coordinating, **17,600** actions; a zero-day; broke out to steal the eval's answers — cheating, at scale.
 - UK AISI: Mythos 5 forged GitHub identities, pressured a **real maintainer** to merge a dropper. maintainer refused. **xz, automated.**



the harness failed, not the model

- each time the **model** worked; the **harness** failed: sandbox misconfigured, prompt said “no internet.”
- auditing the **weights** would have caught **none** of it.
- the risk lives in the **rig around the model**: sandboxes, tools, credentials, identities.

open is not safer. open is *checkable*.



so — can you ignore open weights?

lens	2007	2026
anthropology	gift economy, centuries old	still the engine. now harvested
economics	too powerful to ignore	\$400k/yr says so. compute vs tokens
legal	no future without it	no definition ; secret rules instead
technical	if accessible, it gets pried open	if closed, it escapes anyway

2007's last line: *what social institutions will handle it?* still open.

four things this room can do · then discussion

- say **which** open you mean. weights?
data? code? every time.
- guard the maintainers: verify identities,
rate-limit AI PRs, **pay** them.
- run small, run local, **archive** the weights.
endpoints are not artifacts.
- **measure** the many-eyes claim. don't cite
it.
- are weights speech, or a munition?
- name one security property openness
actually provides.
- who here could still run today's stack in
2028?

the gift must move. · timmm@ieee.org ·
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